

Deepansh Sabharwal

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Profile

Third-year Computer Engineering student who works across the stack: real-time firmware in C, a fail-safe controller taken from Verilog to a physical FPGA, a detection model quantized onto an NPU and deployed as a self-recovering service, and a live web platform on TypeScript and PostgreSQL that I run alone. Currently doing supervised research in analog signal correction. Git on every project, Linux daily.

Canadian permanent resident. Available from January 2027 for 4 to 16 months, anywhere in Canada.

Experience

University of British Columbia | Undergraduate Research Assistant **May 2026 - Present**
Embedded systems, edge ML, signal processing / Python, C, embedded Linux, Hailo NPU, LTspice, SDR

- Currently designing a polynomial predistortion stage to cancel cubic distortion in an amplifier chain, modelled in LTspice with coefficients fit in Python from swept measurements and an SDR feedback loop that re-fits them as conditions drift
- Built and deployed a real-time wildlife detection system: fine-tuned YOLOv8n (mAP50 0.823), quantized to INT8 for the NPU, and root-caused a compiler partition failure rather than working around it; runs as a systemd service verified by cold boot to detect within about 45 seconds of power-on
- Fixed failures that only appeared on the physical unit (a signal handler overridden by a vendored library, a thread teardown race confirmed with a 20-trial stress test); maintain 47 unit tests behind a simulation backend

Janvani, Citizen Journalism Platform | Founder and Sole Developer **2026 - Present**
Next.js, React, TypeScript, Node.js, PostgreSQL, Redis, GitHub Actions CI / Live: janvani-sable.vercel.app

- Designed the schema and REST API, built the front end, and ship every change through automated CI with production monitoring; ran a 16-point security review before launch and fixed an access-control flaw and stored secrets it found

Best Buy Canada | Sales Advisor **Jul 2025 - Present**
Kelowna, BC. Customer-facing technical role, 20+ consultations daily, explaining technical options in plain language

Projects

RISC-V Toolchain to Hardware Emulation Pipeline | Solo **2025**
C, RISC-V Assembly, gdb, Linux

- Wrote a cycle-accurate model of a five-stage pipelined processor in C with forwarding and stalling as separate modules, plus the cross-compiler that targets it; debugged hazard defects in gdb to about 1.0 IPC against a 25-program regression suite

FaultLatchedPWM | Solo **2026**
Verilog HDL, Tang Nano 20K FPGA / github

- Designed a deterministic PWM controller with a five-state latched fault machine, simulated it, then verified it on a physical board through synthesis and place-and-route

AI-in-Farming: Crop Disease Classifier | Solo **2025**
Python, TensorFlow, NumPy, pandas. Convolutional classifier on about 15,000 leaf images, 92% test accuracy, improved through confusion analysis

Technical Skills

Languages: C (strong), Python (strong), TypeScript, JavaScript, Java, SQL, Bash, Verilog HDL, RISC-V Assembly
Embedded and hardware: Real-time firmware, embedded Linux, systemd, UART / I2C / SPI / GPIO, FPGA design flow, LTspice, software-defined radio, gdb, GCC cross-compiler
ML and data: TensorFlow, YOLOv8, OpenCV, INT8 quantization, model evaluation, pandas, PostgreSQL schema design
Web: React, Next.js, Node.js, REST APIs, Redis, GitHub Actions CI/CD, production monitoring, security review
Practice: Root cause debugging, test infrastructure with simulation backends, technical documentation, Claude Code and Copilot as daily coding agents

Education

University of British Columbia, Okanagan | B.A.Sc. Computer Engineering, expected May 2029 Kelowna, BC
Coursework: Microcomputer Engineering, Microelectronics I, Linear Circuit Theory, Industrial Automation, Data Structures and Algorithms (Java)